Manual

Lab1: Bug Report Classification – ISE Coursework

Overview
This baseline tool automatically classifies a bug report from five deep learning projects Caffe, Incubator-mxnet, Keras, PyTorch and Tensorflow as performance-related (1) or not (0). Using TF-IDF features, it compares three models: Gaussian Naive Bayes (baseline), Linear SVM, and Random Forest.

Setup

1. Clone or download the project repository
2. Navigate to the project directory in your terminal

    ISE_COURSEWORK_LAB1

3. Create and activate a virtual environment:

    python3 -m venv venv
    source venv/bin/activate

4. Install required dependencies

    pip install -r requirements.txt

Dataset Setup
The datasets used in this project are five CSV files containing GitHub bug reports from the following deep learning projects: Caffe, Incubator-mxnet, Keras, PyTorch and Tensorflow.

These files can be downloaded from: https://github.com/ideas-labo/ise-lab-solution/tree/main/lab1

Once downloaded, place all five CSV files inside the data/ folder in the root directory as shown below:

```
ISE_COURSEWORK_LAB1/
  main.py
  requirements.txt
  data/
    caffe.csv
    incubator-mxnet.csv
    keras.csv
    pytorch.csv
    tensorflow.csv
```

The script will automatically locate and process all five files when run.

Running Tool
Ensure the virtual environment is active, then run:

    python3 main.py

The tool will automatically process all five projects in sequence. No additional arguments or configuration are required.

Understanding the Output
For each project, the tool prints a results table showing the average performance of each model across 30 repeated experiments:

    Model Acc Prec Rec F1-Mac F1-Bin AUC

The columns are:
Acc: Accuracy - overall proportion of correct predictions

Prec: Precision (macro) - ability to avoid false positives
Rec: Recall (macro) - ability to find all positive cases
F1-Mac: Macro F1 - harmonic mean of macro precision and recall
F1-Bin: Binary F1 - F1 score for the positive class (performance bugs) only
AUC: Area Under the ROC Curve - overall discrimination ability

Following the results table, statistical analysis is printed showing the Wilcoxon Signed-Rank Test p-value and Cliff's Delta effect size for each proposed model compared against the baseline.